信 息 工 程 学 院

《大数据采集与清洗》实验报告

学院：信息工程学院　**班级**：数科 2102　**姓名**：陈　**学号**:202101　**成绩**:

一、实验题目

Kafka 快速开始（Python 版）

二、实验目的

掌握 Kafka 快速上手

三、实验平台

希冀平台

Kafka 3.5.1

四、实验过程及步骤

1. 解压 Kafka

```
USER_ID: 0, GROUP_ID: 0
(base) root@cg:~# rm -rf /tmp/kafka-logs /tmp/zookeeper
(base) root@cg:~# cd /usr/local/support/
(base) root@cg:/usr/local/support# tar -xzf kafka_2.12-3.5.1.tgz
(base) root@cg:/usr/local/support# cd kafka_2.12-3.5.1
(base) root@cg:/usr/local/support/kafka_2.12-3.5.1# ls
bin  config  libs  LICENSE  licenses  NOTICE  site-docs
(base) root@cg:/usr/local/support/kafka_2.12-3.5.1#
```

2. 启动 Kafka

```
(base) root@cg:/usr/local/support/kafka_2.12-3.5.1# bin/zookeeper-server-start.sh config/zookeeper.properties
[2023-11-25 00:43:28,539] INFO Reading configuration from: config/zookeeper.properties (org.apache.zookeeper.server.qu
rum.QuorumPeerConfig)
[2023-11-25 00:43:28,540] WARN config/zookeeper.properties is relative. Prepend ./ to indicate that you're sure! (org.
pache.zookeeper.server.quorum.QuorumPeerConfig)
[2023-11-25 00:43:28,543] INFO clientPortAddress is 0.0.0.0:2181 (org.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2023-11-25 00:43:28,543] INFO secureClientPort is not set (org.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2023-11-25 00:43:28,543] INFO observerMasterPort is not set (org.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2023-11-25 00:43:28,543] INFO metricsProvider.className is org.apache.zookeeper.metrics.impl.DefaultMetricsProvider (
rg.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2023-11-25 00:43:28,545] INFO autopurge.snapRetainCount set to 3 (org.apache.zookeeper.server.DatadirCleanupManager)
[2023-11-25 00:43:28,545] INFO autopurge.purgeInterval set to 0 (org.apache.zookeeper.server.DatadirCleanupManager)
[2023-11-25 00:43:28,545] INFO Purge task is not scheduled. (org.apache.zookeeper.server.DatadirCleanupManager)
[2023-11-25 00:43:28,545] WARN Either no config or no quorum defined in config, running in standalone mode (org.apache
zookeeper.server.quorum.QuorumPeerMain)
[2023-11-25 00:43:28,546] INFO Log4j 1.2 jmx support not found; jmx disabled. (org.apache.zookeeper.jmx.ManagedUtil)
[2023-11-25 00:43:28,547] INFO Reading configuration from: config/zookeeper.properties (org.apache.zookeeper.server.qu
rum.QuorumPeerConfig)
[2023-11-25 00:43:28,547] WARN config/zookeeper.properties is relative. Prepend ./ to indicate that you're sure! (org.
pache.zookeeper.server.quorum.QuorumPeerConfig)
[2023-11-25 00:43:28,547] INFO clientPortAddress is 0.0.0.0:2181 (org.apache.zookeeper.server.quorum.QuorumPeerConfig)
```

实验报告的内容与格式按任课教师的要求书写。

```
(base) root@cg:/usr/local/support/kafka_2.12-3.5.1# bin/kafka-server-start.sh config/server.properties
[2023-11-25 00:47:55,386] INFO Registered kafka:type=kafka.Log4jController MBean (kafka.utils.Log4jControllerRegistrati
on$)
[2023-11-25 00:47:55,812] INFO Setting -D jdk.tls.rejectClientInitiatedRenegotiation=true to disable client-initiated T
LS renegotiation (org.apache.zookeeper.common.X509Util)
[2023-11-25 00:47:55,875] INFO Registered signal handlers for TERM, INT, HUP (org.apache.kafka.common.utils.LoggingSign
alHandler)
[2023-11-25 00:47:55,876] INFO starting (kafka.server.KafkaServer)
[2023-11-25 00:47:55,877] INFO Connecting to zookeeper on localhost:2181 (kafka.server.KafkaServer)
[2023-11-25 00:47:55,905] INFO [ZooKeeperClient Kafka server] Initializing a new session to localhost:2181. (kafka.zook
eeper.ZooKeeperClient)
[2023-11-25 00:47:55,911] INFO Client environment:zookeeper.version=3.6.4--d65253dcf68e9097c6e95a126463fd5fdeb4521c, bu
ilt on 12/18/2022 18:10 GMT (org.apache.zookeeper.ZooKeeper)
[2023-11-25 00:47:55,911] INFO Client environment:host.name=cg (org.apache.zookeeper.ZooKeeper)
[2023-11-25 00:47:55,911] INFO Client environment:java.version=11.0.19 (org.apache.zookeeper.ZooKeeper)
[2023-11-25 00:47:55,912] INFO Client environment:java.vendor=Ubuntu (org.apache.zookeeper.ZooKeeper)
[2023-11-25 00:47:55,912] INFO Client environment:java.home=/usr/lib/jvm/java-11-openjdk-amd64 (org.apache.zookeeper.Zo
oKeeper)
[2023-11-25 00:47:55,912] INFO Client environment:java.class.path=/usr/local/support/kafka_2.12-3.5.1/bin/../libs/activ
ation-1.1.1.jar:/usr/local/support/kafka_2.12-3.5.1/bin/../libs/aopalliance-repackaged-2.6.1.jar:/usr/local/support/kaf
ka_2.12-3.5.1/bin/../libs/argparse4j-0.7.0.jar:/usr/local/support/kafka_2.12-3.5.1/bin/../libs/audience-annotations-0.1
3.0.jar:/usr/local/support/kafka_2.12-3.5.1/bin/../libs/commons-cli-1.4.jar:/usr/local/support/kafka_2.12-3.5.1/bin/../
libs/commons-lang3-3.8.1.jar:/usr/local/support/kafka_2.12-3.5.1/bin/../libs/connect-api-3.5.1.jar:/usr/local/support/k
```

3. 创建 Topic 存放消息

```
(base) root@cg:~# cd /usr/local/support/kafka_2.12-3.5.1
(base) root@cg:/usr/local/support/kafka_2.12-3.5.1# bin/kafka-topics.sh --create --topic quickstart-events --bootstrap-
server localhost:9092
Created topic quickstart-events.
(base) root@cg:/usr/local/support/kafka_2.12-3.5.1# bin/kafka-topics.sh --describe --topic quickstart-events --bootstra
p-server localhost:9092
Topic: quickstart-events        TopicId: GeYKgvQyQq-bcnqQR-fd9Q PartitionCount: 1       ReplicationFactor: 1    Configs
:
        Topic: quickstart-events        Partition: 0    Leader: 0       Replicas: 0     Isr: 0
(base) root@cg:/usr/local/support/kafka_2.12-3.5.1#
```

4. 创建消息（事件）到指定的 Topic

```
        Topic: quickstart-events        Partition: 0    Leader: 0       Replicas: 0     Isr: 0
(base) root@cg:/usr/local/support/kafka_2.12-3.5.1# bin/kafka-console-producer.sh --topic quickstart-events --bootstrap
-server localhost:9092
>This is my first event
>This is my second event
>Wo yao wan Genshin
>
```

5. 读取消息（事件）

```
(base) root@cg:~# cd /usr/local/support/kafka_2.12-3.5.1
(base) root@cg:/usr/local/support/kafka_2.12-3.5.1# bin/kafka-console-consumer.sh --topic quickstart-events --from-begi
nning --bootstrap-server localhost:9092
This is my first event
This is my second event
Wo yao wan Genshin
```